Ex. No. 12	BUILD WEB APP WITH FLUTTER
Date of Exercise	
Git Hub Host Link	https://github.com/gurubruhh/Flutter-Laboratory.git

Aim

Description

This experiment focuses on creating a responsive web application using Flutter. The application is executed through a web browser such as Chrome. LayoutBuilder is used to adjust the interface according to different screen sizes. Finally, a production web build is generated for deployment.

Procedure:

1. Check available web devices using flutter devices.
2. Create a project using flutter create flutter_web_app.
3. Navigate into the project directory.
4. Open the project using code ..
5. Design the web interface using Scaffold and AppBar.
6. Add an ElevatedButton with a Welcome message.
7. Use LayoutBuilder to implement responsive layouts.
8. Run the application using flutter run -d chrome.
9. Resize the browser window and test mobile and desktop layouts.
10. Generate production files using flutter build web.

Program

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Flutter Web Application',
      theme: ThemeData(
```

```
primarySwatch: Colors.blue,
      ),
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Flutter Web Application'),
        backgroundColor: Colors.blue,
        foregroundColor: Colors.white,
      ),

      body: LayoutBuilder(
        builder: (context, constraints) {
          bool isMobile = constraints.maxWidth < 600;

          return Center(
            child: Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                Icon(
                  Icons.web,
                  size: isMobile ? 70 : 100,
                  color: Colors.blue,
                ),

                const SizedBox(height: 20),

                Text(
                  'Welcome to Flutter Web!',
                  textAlign: TextAlign.center,
                  style: TextStyle(
                    fontSize: isMobile ? 22 : 30,
                    fontWeight: FontWeight.bold,
                  ),
                ),

                const SizedBox(height: 20),

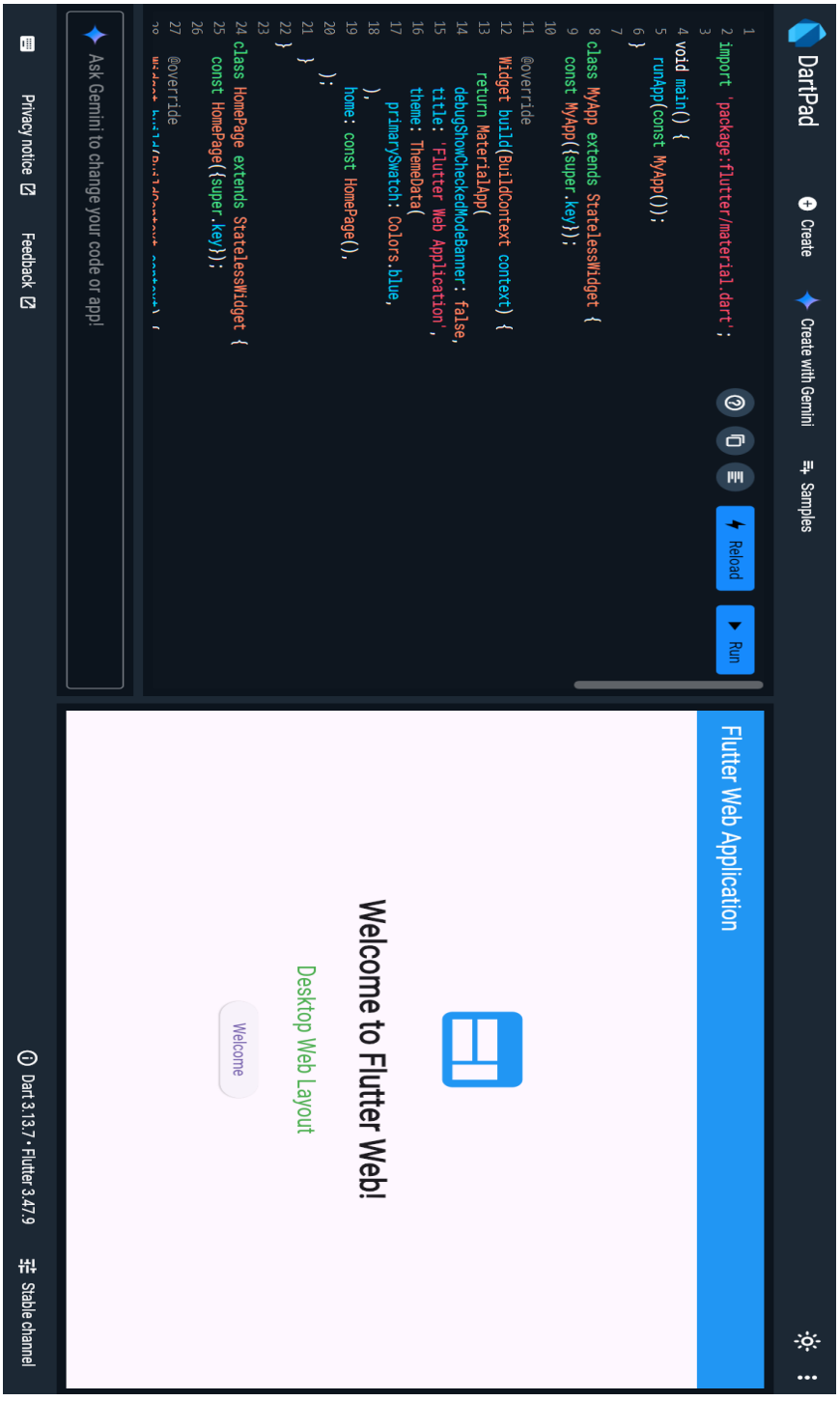
                Text(
                  isMobile
                      ? 'Mobile Web Layout'
```

```
'Desktop Web Layout',
                style: const TextStyle(
                  fontSize: 20,
                  color: Colors.green,
                ),
              ),

              const SizedBox(height: 25),

              ElevatedButton(
                onPressed: () {
                  ScaffoldMessenger.of(context).showSnackBar(
                    const SnackBar(
                      content: Text('Welcome Button Clicked!'),
                    ),
                  );
                },
                child: const Text('Welcome'),
              ),
            ],
          ),
        );
      },
    ),
  );
 }
}
```

Output


DartPad
Create
Create with Gemini
Samples
Reload
Run
import 'package:flutter/material.dart';
void main() {
runApp(const MyApp());
}
class MyApp extends StatelessWidget {
const MyApp({super.key});
@override
Widget build(BuildContext context) {
return MaterialApp(
debugShowCheckedModeBanner: false,
title: 'Flutter Web Application',
theme: ThemeData(
primarySwatch: Colors.blue,
),
home: const HomePage(),
);
}
}
class HomePage extends StatelessWidget {
const HomePage({super.key});
@override
Ask Gemini to change your code or app!
Flutter Web Application
Welcome to Flutter Web!
Desktop Web Layout
Welcome
Privacy notice
Feedback
Dart 3.13.7 • Flutter 3.47.9
Stable channel

Result

Thus, the Flutter applications were successfully developed, executed, and tested on the respective platforms. The required functionalities, user interface, and responsive layouts were verified successfully.